

Workshop on Multinomial-Processing-Tree (MPT) Modeling

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Day 1: Essentials of MPT Modeling

- 9:00 – 10:30 Basics
 - Introduction to standard MPT models (logic, examples, advantages, limitations)
 - Model development (model construction, paradigm, data structure, identifiability)
 - Parameter estimation (maximum likelihood, EM algorithm)
 - Model assessment (goodness-of-fit statistics: G^2 , Pearson's χ^2 , and power divergence PD^2)
- Break
- 11:00 – 12:15 Application I: Basic use of multiTree
 - EQN syntax, data files, parameter estimation, model fit
 - Identifiability concepts and checks
 - Practical exercises
- Noon Break
- 13:30 – 14:45 Advanced features of multiTree
 - Order constraints & testing interactions
 - A-priori and post-hoc power analyses
 - Power optimization
- Break
- 15:15 – 16:30 Application II: Developing a new MPT model
 - Specification of experimental paradigm, data structure, and parameters
 - Workflow with multiTree: model specification and testing

Day 2: Advances in MPT Modeling

- 9:00 – 10:30 Bayesian hierarchical MPT modeling
 - MPT models & heterogeneity
 - Hierarchical MPT models
 - Bayesian estimation with MCMC sampling
 - Correlation and regression with continuous covariates
- Break
- 11:00 – 12:15 Application III: TreeBUGS
 - Practical exercises on hierarchical MPT modeling using TreeBUGS in R
 - Basics: Model fitting, convergence, plots, model fit
 - Advanced topics: within-/between-subject comparisons, covariates, simulations
- Noon Break
- 13:30 – 14:30 Modeling continuous data using mixture models
 - Modeling response times with histograms (MPT-RT)
 - Parametric modeling with generalized processing trees (GPT)
 - Serial process model for response times (RT-MPT)
- Break
- 15:00 - 16:00 Application IV
 - Questions and answers
 - Developing and testing (new) models suggested by the participants